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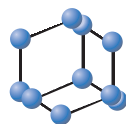
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REVIEW ARTICLE

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Application of Traditional Chinese Herbal Medicine By-products as Dietary Feed Supplements and Antibiotic Replacements in Animal Production



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Abstract: Background: Misuse of synthetic antibiotics in livestock leads to the transfer of antibiotic resistant pathogens into humans and deposits toxic residues in meat and milk. There is therefore an urgent need for safe and viable alternative approaches to improve the nutrition and wellbeing of farm animals. An alternative source that has been widely exploited is Traditional Chinese Herbal Medicine (TCHM). These herbs contain several but less toxic bioactive compounds which are generally regarded as biodegradable. Recently, advances in the knowledge of the importance of TCHM have led to a rapid increase in its production and hence, increasing the amount of by-products generated. Such by-products have become a serious environmental challenge because producers regard them as industrial waste and discard them directly. This review summarizes scientific findings on the bioactive compounds in TCHM and TCHM by-products, discusses functional dietary patterns and outlines challenges that may hinder full utilization of TCHM by-products in animal production.

Methods: Information for this review was obtained through scientific databases and websites such as Pubmed and Google scholar from 2004 to 2017 using experimental studies on bioactive compounds in TCHM and their effects in animal production.

Results: Studies have shown that TCHM by-products contain high amounts of bioactive compounds which confer several nutritional and health benefits to animals and thus could be incorporated as feed additives.

Conclusion: The findings for this review indicate that TCHM by-products apart from being a good alternative for synthetic antibiotics could also minimize the current environmental challenges associated with its disposal.

Keywords: Animal production, antibiotics, antioxidants, bioactive compounds, environmental pollution, traditional Chinese herbal medicine.

1. INTRODUCTION

World population is gradually becoming more urbanized, and so the demand for meat and meat products is anticipated to rise in the coming years [1, 2]. Many countries are, however, facing animal health issues that affect economic resources and reduce productivity [3]. The commonest therapy in animal healthcare is the use of synthetic antibiotics which refer to a wide variety of compounds with antibiotic activities [3]. Synthetic antibiotics can be administered to farm animals to prevent and treat infectious diseases and to improve animal productivity [4]. For decades, synthetic antibiotics have been successfully included in livestock feed as growth promoters to improve animal performance [5]. However, overuse of antibiotics may transfer antibiotic-resistant pathogens to humans and also deposit toxic residues in meat and milk [5, 6]. Furthermore, public recognition of feed supplementation with antibiotics is facing a decline for the same reason [7]. Because of these concerns, many countries in Europe have placed legal restrictions on antibiotic supplementation in animal feed [8]. Efforts are therefore being made to find safe and effective alternative medicinal options for livestock around the world [8].

Traditional Chinese Herbal Medicine (TCHM) plants such as *Astragalus membranaceus*, cowherb seed (*Vaccaria hispanica* seed), *Angelica sinensis*, *Licorice rhizome*, *Ligusticum wallichii* and *Codonopsis pilosula* have been shown to improve feed efficiency, growth performance and immunity, and may inhibit oxidative, bacterial and viral activities as antibiotic replacements [6, 8, 9]. Secondary metabolites in TCHMs are extracted through decoction or other advanced methods. The efficiency of extraction is however

about 50%, indicating that by-products still contain high amounts of bioactive compounds [10]. For instance, by-products of Danshen (common Chinese name for *Salvia miltiorrhiza* Bunge roots) constitute more than 80% of the feedstock weight. A large amount of the by-products are therefore produced annually [11]. Danshen is well-known for the treatment of cardiovascular and cerebrovascular diseases [11]. The annual by-product yield in China is approximately 30 million tons, disposal of which is becoming a serious environmental challenge [12]. Producers regard TCHM by-products as industrial waste and discard them mainly by piling, burying or burning, resulting not only in environmental pollution but also resource wastage [13].

There are limited reports on the reuse of TCHM by-products in livestock production. For example, Artichoke by-products have been used for silage production because of its good odour, fodder, crude protein (88 g/kg DM) and fat (509 g/kg DM) contents [14]. Artichoke (*Cynara scolymus* L.) by-product silage has, therefore, been used to replace straw in ruminant feeding [15]. Artichoke fodder showed no significant difference in milk composition when it was fed to dairy cows, but recorded lower fat and total free fatty acids in manufactured cheese [16].

This review summarizes related findings on the dietary use of TCHM by-products in animal feeding and discusses their effect in livestock production. Based on the assertion that TCHM by-products contain a high amount of bioactive compounds [10], which are not different from those of the main herbal plants, we have also discussed research findings on the dietary effects of some TCHMs whose by-products could serve as feed additives.

2. CONCEPT OF TCHM

With a long history of medical practice, Traditional Chinese Medicine (TCM) has developed with outstanding hypothetical structure and profound medical experience which is used to im-

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prove human and animal health. TCM theory is based on “holistic concept” of philosophy which relies on syndrome-based treatments and multi-targeting/multicomponent modes of action and thus ensures an inimitable advantage in the treatment of complex diseases or disorders [17]. TCM modules include acupuncture, TCHM, moxibustion and t'ai chi. TCHM is one of the most commonly employed therapies in TCM [18] and thus this review focuses on the functional uses of TCHM by-products. TCHM has been used for more than 2000 years in China to improve animal health and production [17] and more recently, to modulate nutritional metabolism [19, 20], immune responses [21, 22], and intestinal health [23] of livestock. TCHM, therefore, provides a striking alternative to dietary antibiotics in livestock production. The effect of TCHM in improving animal growth performance is parallel to that of dietary antibiotics, and so some commercial products have been developed from these formulated herbs and are extensively used in the commercial production of poultry and swine in China [17].

3. BIOACTIVE COMPOUNDS IN TCHM BY-PRODUCTS

A bioactive compound is a compound with the ability to interact with a part or some parts of the living tissue by presenting a wide range of probable effects [24]. The sources of these substances can be natural (plants, animals or other sources) or synthetic (partially or totally). Studies have been done on the roles (Table 1) of different bioactive compounds [25] in animals. Recently, great attention is being paid to the retrieval of bioactive components from a variety of plant materials and industrial by-products including seeds, grain, leaves, pulp, peels, culls, hulls, rinds, malts, barriers, marc, pits, press, cakes, hops, husks, wood, spent, among others [26]. Herbs synthesize chemical compounds during regular metabolic activities [27, 28]. Such compounds can be categorized into primary (sugars and oils) or secondary compounds (phytochemicals) and when administered, influence the wellbeing of animals [28]. Phytochemicals comprise medicinal herbs which are valued for their flavour and medicinal properties and may be further categorized according to their healing properties or according to their mode of preparation (decoction infusion tincture, maceration, syrup and inhalation) [28]. Different animals are treated with herbal remedies worldwide (31% cattle, 14% horses, 17% sheep, 17% goats, 7% pigs, 9.1% poultry, 5.3% dogs, and 4.3% rabbits) mainly due to their efficacy against pests and diseases [28]. Therefore, there has been an increase in the use of TCHM extracts in livestock production as appetizers, antioxidants, and colourants, among other functions [28, 29].

Lipid oxidation and auto-oxidation are the main agents that deteriorate meat quality and are prevalent during storage and processing [30]. Lipid oxidation affects colour, texture, odour, flavour and nutritional value of meat by developing bad odours, off-flavours and toxic compounds (aldehydes and peroxides) harmful to human health [30, 31]. Beef is prone to lipid oxidation because it contains highly unsaturated fatty acids [30]. One of the main bioactive functions of TCHM and TCHM by-products is their antioxidant abilities (Table 1) because they contain compounds such as flavonoids, phenolic acids, tocopherols and carotenoids [32]. Antioxidants are integrated within the cell membranes of target organisms and attack oxygen species that promote oxidative activities, thereby preserving meat quality [32]. Synthetic antioxidants that have been commonly used include butylated hydroxytoluene, butylated hydroxyanisole, propyl gallate and tertiary butyl hydroquinone [30]. Carcinogenicity potentials and residual deposits of such compounds, however, make them unfavourable for use. Bioactive compounds in TCHM and related dietary functions (antifungal, antibacterial, anticancer, anti-inflammatory, antiviral, immune stimulants, etc.) are depicted in Table 1. Chemical structures of TCHM bioactive compounds are presented in Fig. (1).

4. APPLICATION OF TCHM BY-PRODUCTS IN ANIMAL PRODUCTION

A wide range of herbs, spices and other phytogetic products have been used extensively in recent years as substitute antibiotics because of their natural plant-derived properties and ability to improve growth performance [33]. However, TCHM and TCHM by-products should be used cautiously due to possible side effects in animals [34], despite several studies that have shown that they have positive effects on growth performance and meat quality in animals and are less toxic. For example, supplementing animal diets with antioxidant based phytochemicals yields meat and meat products that contain antioxidants and thus improve meat quality [35]. Also, the colour of beef improves with the addition of antioxidants and results in lower absorption of iron [34].

Technologies related to the application of TCHM by-products in animal production involves either full or partial replacement of some feed ingredients in animal basal diet with the by-product [8, 36] and in some cases, merely adding the by-product to the basal diet without any alterations [37, 38] based on the type and chemical composition of the by-product used. Generally, by-products are dried to constant weight and ground into powder before supplementation [32]. For non-ruminants, by-products could be fermented before supplementation. Fermentation reduces the fibre content of by-products by over 50% and improves nutritional composition [8]. Recent findings on the effects of TCHM by-products in pig, poultry and ruminant production are elaborated herewith.

4.1. Application in Swine

Several studies have indicated that TCHM offers a wide range of benefits in pig production. *Saururus chinensis*, also known as Chinese lizard's tail is a wild perennial crop used in TCHM for treating diuretic and other diseases, such as gonorrhoea, oedema, jaundice and inflammatory diseases [39]. Main active compounds in *S. chinensis* include flavonoids, alkaloids, essential oils, quercetin, neolignans, lignans, quercitrin, rutin and tannins [39]. Although growth performance and meat quality of pigs did not improve when finishing pig diet was supplemented with 1 and 2 g/kg *S. chinensis* extract, noxious gas (NH₃) emission was significantly reduced ($P < 0.05$) in the 2 g/kg treatment on days 0 and 10 [39]. Similarly, TCHM mixture (ginger, buckwheat, thyme, Curcuma and black pepper) had no significant effect on Average Daily Gain (ADG), Average Daily Feed Intake (ADFI) and Feed to Gain (F:G) ratio in weanling pigs. Nutrient digestibility was however improved as Dry Matter (DM), and nitrogen (N) digestibility increased on days 7 and 42 [40]. Also, TCHM formula (*Radix Astragali* root, *Radix Glycyrrhizae* root, *Cortex Eucommiae* leaf, *Rhizoma Atractylodis Macrocephalae* rootstalk, *Rhizoma Zingiberis* rootstalk, *Fructus Crataegi* fruit, *Fructus Rosae Laevigatae* fruit, *Fructus Ligustri Lucidi* fruit and *Pericarpium Citri Reticulatae* pericarp) had no effect on growth performance (body weight (BW), ADG, ADFI and F:G ratio), carcass characteristics (Hot carcass weight, carcass straight length, oblique length, skin thickness, backfat depth, loin eye area, carcass lean percentages, dressing percentage, left ham weight, ham percentage, heart weight, liver weight, spleen weight, lung weight and kidney weight) and some meat quality parameters (cooking loss, drip loss, shear force, pH, moisture and CP) of finishing pigs. Conversely, crude fat content increased, and Thiobarbituric Acid Reactive Substances (TBARS) values were significantly lower compared with control diet. Meat colour and marbling scores were also improved [41]. On the contrary, a study conducted on the effect of fermented TCHM (*Gynura procumbens*, *Rehmannia glutinosa* and *Scutellaria baicalensis*) on weanling pigs revealed that TCHM improved growth performance (ADG, ADFI and F:G ratio) of pigs. Furthermore, Apparent Total Tract Digestibility (ATTD) of DM, N and Gross Energy (GE); noxious gas emission (ammonia, hydrogen sulfide and total mercaptans); and diarrhea

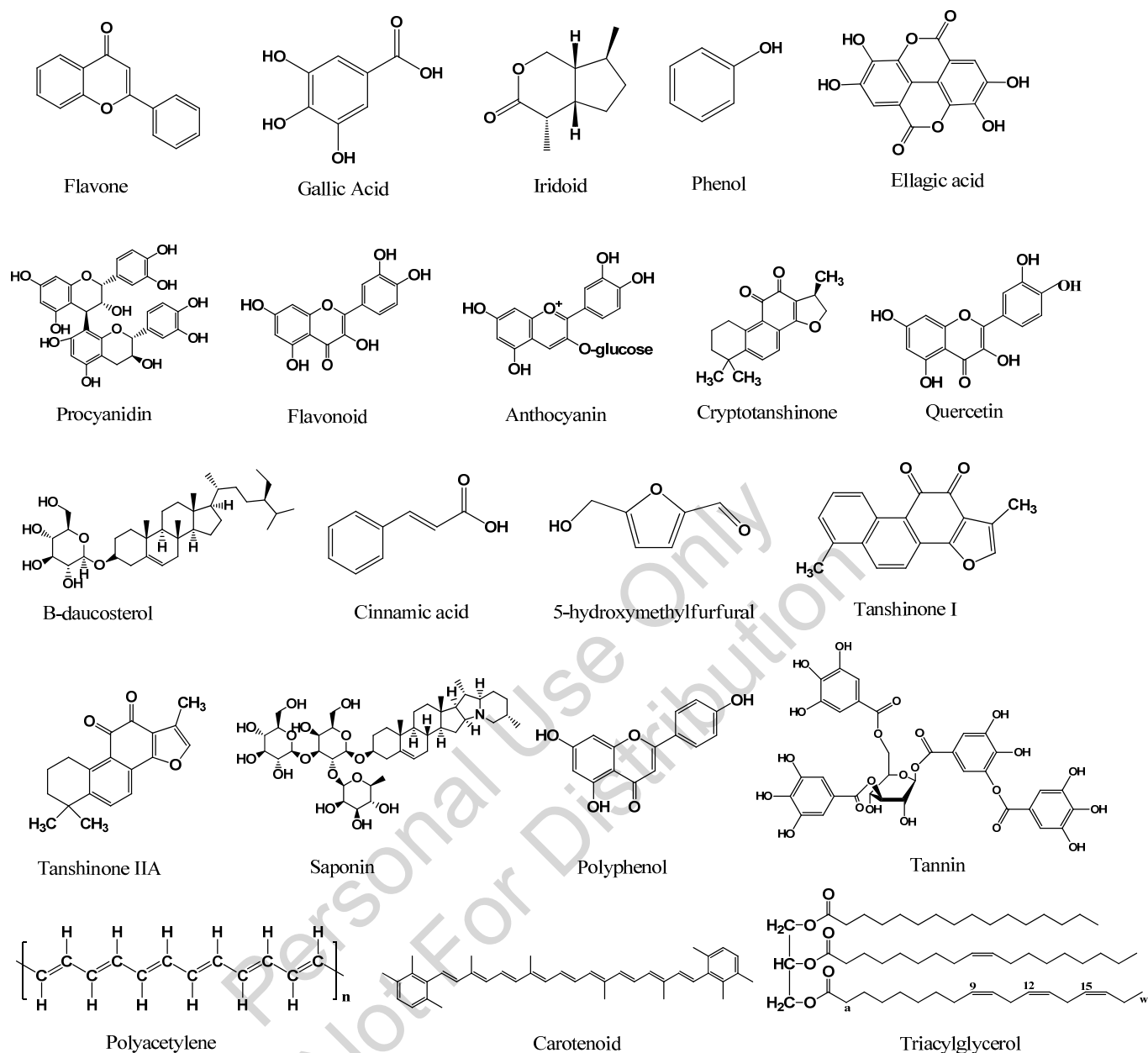


Fig. (1). Chemical structures of TCHM bioactive compounds.

score in pigs were improved in the same study [33]. *Coptis chinensis*, a known TCHM, is a good source of berberine which is known for its numerous antioxidant and antibiotic activities [42]. Growth performance (BW, ADG and ADFI) and ATTD of DM, N and organic matter (OM) in growing-finishing pigs were not affected ($P > 0.05$) by dietary supplementation of *C. chinensis* extract. Meat quality, however, had better ($P < 0.05$) pH (6.02), meat colour (6.26) and Water Holding Capacity (WHC), (54.1) in pigs fed 1.0 g/kg of the extract than pigs fed with the control diet (pH, 5.74; meat colour, 8.20; WHC, 49.3) [42].

Despite the numerous studies on TCHM, only a few studies have been done on the effects of TCHM by-products in animal production. Some studies revealed that by-products have positive effects on growth performance and meat quality in pigs [43]. Results of the findings on the effects of TCHM by-products in pigs are summarized in Table 2. *Ginkgo biloba* L. by-products contain many bioactive compounds including proteins, flavonoids, terpenoids,

polysaccharides and mineral substances and have been reported to offer beneficial effects in piglets [8]. The addition of 0.5% each of green tea (*Camellia sinensis*) by-products and green tea probiotics improved growth performance and immunity (Table 2) of pigs [44]. For centuries, the Chinese, Japanese and Koreans have used green tea as an anti-ageing herb [44]. The use of green tea in animal production is however costly compared with other TCHMs. Therefore, it is more economical to use green tea by-products [44]. The by-products are generated from green tea beverage drink production and are known for its economical and efficient use especially as feed supplements [44].

Based on the findings collated, growth performance of pigs is not greatly influenced by TCHM by-products. This could be due to pigs' inability to digest some TCHM by-products since they contain high amounts of fibre. However, nutrient digestibility, immunity and meat quality in pigs have been significantly influenced by TCHM and TCHM by-product supplementation.

4.2. Application in Poultry

The worldwide poultry industry has been responding to the increasing demand for meat as World population is anticipated to rise to more than 9 billion by 2050 [45, 46]. To maintain this demand, production efficiency must be kept at a high peak. Several studies have been carried out on the potential benefits of TCHM in poultry with a few on TCHM by-products.

Sanguinarine is an active component of *Macleaya cordata* and is known to exhibit anti-inflammatory, anti-microbial, analgesic and immunomodulatory activities when used in TCHM [46]. Growth performance, Weight Gain (WG), Feed Intake (FI) and Feed Conversion Ratio (FCR) at days 22-35 improved significantly against control group when broiler diets were supplemented with 50 ppm Sanguinarine. Also, 20 and 50 ppm Sanguinarine supplementation significantly lowered serum total cholesterol and Malonaldehyde (MDA) levels in meat [46]. Similarly, WG and FCR in stimulated Taiwan country chicken fed 0.3% TCHM complex (15% *Glycyrrhiza glabra*, 15% *Rosmarinus officinalis*, 25% *Salvia officinalis*, 20% *Lavandula angustifolia* and 25% *Ocimum basilicum*) improved significantly. The TCHM complex also improved immunity as serum γ -globulin levels increased. On carcass and meat quality, colour scores in both cases improved, possibly due to antioxidants polyphenols and flavonoids in the herbs. Conversely, 0.5 and 0.7% of the complex supplemented group recorded better WHC. Also, taste, flavour, juiciness and overall-acceptance of thigh muscle meat improved in the TCHM supplemental group when subjected to sensory evaluation [47]. Studies on the effect of TCHM, wild-ginseng adventitious root meal in broiler chickens revealed that, growth performance (body weight gain (BWG) and Feed Intake (FI)) in the TCHM supplemented group were improved [48]. TCHM also increased lymphocyte count and decreased total cholesterol concentration. Again, TBARS values in the TCHM supplement group decreased significantly, indicating antioxidant activity on broiler meat. In contrast, there was no significant effect on growth performance (WG, FI, and FCR) when chicken diet was supplemented with Bazhen as a TCHM. Immunity was however improved as Serum Red Blood Cells (SRBC), Newcastle Disease (ND) titers, immunoglobulin G (IgG), and γ -globulin levels increased in TCHM group. Also, catalase activity and level of scavenging α - α -diphenyl- β -picrylhydrazyl (DPPH) radicals increased while the TBARS levels and total glutathione decreased in TCHM group. Again, the TCHM group recorded lower pH and Thiobarbituric Acid (TBA) values and had better color scores than the control group in the breast muscle of chicken [49]. Bazhen is a TCHM comprising eight different kinds of herbs, including *Atractylodes ovate*, *Codonopsis pilosula*, *Wolfiporia Extensa*, *Glycyrrhiza uralensis*, *Angelica Sinensis*, *Ligusticum chuanxiong*, *Paeonia albiflora* Pallas var. *trichocarpa* and *Rehmannia glutinosa*. Phytochemicals in Bazhen include flavonoids, ferulic acid and atractylenolide [50, 51]. Similarly, Korean ginseng [52] and dried tangerine peel extract [53] have been reported to have no effects on the growth performance of poultry, while diet supplementation with Sanguinarine [46]; *Laminaria japonica* [54]; *Saposhnikovia divaricate*, *Lonicera japonica*, and *Chelidonium majus* extracts [55]; and *Ginkgo biloba* leaves [56] have all been reported to improve the meat quality of chicken.

Results of the TCHM by-products effects in poultry are summarized in Table 3. The effect of oriental herbal medicine by-product and methyl sulfonyl methane on growth performance and meat quality of duck was studied by Hwang, *et al.* [43]. The results showed that combination of the two products did not affect growth performance. The product however improved meat quality by raising the concentration of crude protein, fatty acids and sulphur, increased WHC, decreased percentage moisture loss during cold storage, and reduced lipid oxidation (Table 3). Similarly, growth performance of broiler chicken did not improve when diets were supplemented with red ginseng marc. Mortality rate, crude fat and serum cholesterol levels were however decreased along with im-

proved meat quality [37]. Ginseng marc is a fibrous and insoluble by-product that is generated through the production of ginseng extracts [37]. Dietary supplementation with Pomegranate (*Punica granatum* L.) by-product, a popular herbal product used in TCHM did not affect ADG and ADFI in broilers, although there was a significant decrease in FCR (Table 3). Pomegranate by-product however improved meat quality by increasing protein and trace mineral content and decreasing Ether Extract (EE) and cholesterol content of meat as well as reducing lipid oxidation [32]. Recently, there has been increased use of pomegranate because of increased awareness on its health benefits such as antioxidant, anticancer, antimicrobial, antimutagenic and antiviral properties [57, 58].

Similar to pigs, birds are unable to digest fibrous materials which may limit the application of TCHM by-products in poultry. Based on the research findings however, supplementation with TCHM and TCHM by-products in poultry show better and more varying results on performance and product quality. This may be due to the kind of TCHM used, the bioactive compounds contained in them as well as the dosage administered.

4.3. Application in Ruminants

Over the last decades, antibiotics have been included in ruminant diet as growth promoters to improve animal performance and prevent or control diseases [5]. However, previous studies have recorded that TCHM could improve growth performance, feed efficiency and immunity, and could also exert anti-bacterial, anti-viral and anti-oxidative effects in livestock [8] mainly because of the action of bioactive compounds in them.

TCHM (*Astragalus membranaceus* roots, *Angelica Sinensis* roots and *Atractylodes lancea* rhizomes) showed better results than hay mixture when it was fed to sheep as it improved growth performance (increased BWG) significantly [59]. Also, growth performance (FI, ADG and FCR) improved when Holstein calves were fed TCHM complex containing 10% *Glycyrrhiza glabra*, 15% *Rosmarinus officinalis*, 25% *Lavandula angustifolia*, 25% *Ocimum basilicum* and 25% *Salvia officinalis*. In the same experiment, TCHM supplemented group showed better meat quality results than control group as cooking losses and shear force were lower, antioxidant (flavonoid and total phenol) capacity was higher, scavenged DPPH ability showed better potency, and tenderness, flavour, juiciness and overall-acceptance showed better results than control group [60]. Huangqi (*Astragalus membranaceus*) is commonly used in TCHM as an immunomodulating herb, containing compounds such as flavonoids, saponins, polysaccharides, amino acids and other trace elements [61, 62]. Polysaccharides are the main bioactive compounds of *A. membranaceus* root and play key antioxidant roles [63]. Growth performance and apparent digestibility of DM, CP, fat, Ash-free Neutral Detergent Fibre (aNDFom) and Ash-free Acid Detergent Fibre (ADFom) in lambs did not improve when lambs were fed *A. membranaceus* root. Immunity responses however improved as IgM production and lymphocyte proliferation were induced [63]. A study to evaluate fermented *Alisma canaliculatum* Probiotic (ACP) as possible feed additives for finishing Hanwoo (*Bos taurus coreanae*) steers showed interesting results on the experimental animals. Dietary ACP resulted in significant increase in BWG and feed efficiency in contrast with the control ($P < 0.01$). Although moisture, CA and CP were unaffected, CF and cholesterol were lower in 0.5% ACP supplemented group ($P < 0.05$). Also, Fe and Mg in beef increased in 0.5% ACP group ($P < 0.01$) even though Ca was unaffected. Furthermore, 0.5% ACP diet enhanced meat fatty acid profile by decreasing saturated fatty acid content ($P < 0.05$), and increasing unsaturated fatty acid and n-3 polyunsaturated fatty acid content ($P < 0.10$). The authors concluded that dietary ACP had positive effects on growth performance, fatty acid profile, meat composition and oxidative stability in Hanwoo steers [64].

Despite several studies on TCHM, just a few studies have been done on TCHM by-products in ruminants. Findings gathered on the effects of TCHM by-products in ruminants are summarized in Table 4. Growth performance (ADG and FE) improved when diets of Hanwoo steers were supplemented with a mixture of medicinal plant by-products (*Angelica gigas nakai*, *Rehmanniae radix preparata*, *Zingiber officinale roscoe*, *Paeoniae radix*, *Cnidii rhizome*, *Zizyphus jujube*, *Bupleuri radix*, *Citrus nobilis makino*, *Prunus mume* and *Plantago asiatica L.*) at 30 g/kg [34]. A study conducted by Jami, *et al.* [38] on the effect of dietary supplementation of concentrated Pomegranate-Residue Extract (CPE) on *in vivo* digestibility in lactating cows (Table 4) revealed that digestibility of CP, DM, NDF, cellulose, and hemicellulose increased in cows fed 4% CPE [32]. Previous studies have revealed that diet supplementation with pistachio by-product, a food processing by-product with medicinal properties, at increased levels, decreases FI, blood glucose, cholesterol, triglyceride and ruminal Ammonia-Nitrogen (AN) concentration; and increases abomasum AN and blood urea nitrogen in dairy cows and Baloochi sheep [36, 65, 66]. Mahdavi, *et al.* [67] also concluded that increased levels of the by-product had undesirable effects on the growth performance of lamb. The effect of replacing lucerne hay with pistachio by-product on apparent digestibility in sheep was studied by Ghasemi, *et al.* [36]. They concluded that total replacement of lucerne with pistachio by-product improved nutrient digestibility by increasing ($P < 0.05$) the apparent digestibility of N, EE, OM and N retention.

Contrary to non-ruminants, TCHM by-products show more positive effects on performance and product quality in ruminants. This is probably due to ruminants' ability to digest fibrous materials which is characteristic of TCHM by-products.

5. CHALLENGES

TCHM by-products may be considered the best fit alternative feed additives as they have been reported to improve animal performance in various ways. Adoption of TCHM and TCHM by-products in animal production is, however, not devoid of challenges. TCHM theory has been considered abstract, complex, and

difficult to comprehend by Western medical professionals mainly due to cultural and philosophical differences [68]. Additionally, the scarcity of well-controlled clinical trials and periodic manifestations of toxicity in TCHM have hindered its acceptance into mainstream Western medicine [68]. Also, production and processing of TCHM for animal feed are done without adequate medical resources and quality control, and also without proper standards for scientific assessment. It is also challenged with inadequate understanding of the fundamental mechanisms involved in their functions [69]. Furthermore, identification of bioactive compounds in many TCHMs have not yet been fully established [17]. Again, getting regulatory approval for the use of TCHM, especially in western countries where TCHM have not been acknowledged is particularly challenging. This is particularly so because several active compounds are involved in the healing mechanism of TCHM while a proper cause-effect relationship is yet to be established [17]. TCHM is also influenced by varying climatic conditions in different locations which significantly affects its quality and efficacy [70]. Additionally, standardization of the methods for processing TCHM for animal feed has not been established [17]. Manufacturers, therefore, use their conventional processing methods which may cause variations in the end product. Despite the numerous benefits of TCHM in animal production, it is currently not certain if its use has significant economic gains because of limited studies on economic feasibility. Even though 8.86% profit increase was recorded in swine upon the use of TCHM feed additives, no such records have been found for other livestock [17]. Moreover, a full conversion from synthetic antibiotics to the use of TCHM in animal feeding may cause competition between humans and animals for TCHM products [17]. This will stimulate the need to produce more TCHM which will consequently influence the amount of by-product generated. This can be avoided by increasing focus on the use of TCHM by-products for animal feeding while maintaining pure extracts for human use. There is, however the possibility to close the gap between TCM and Western medicine with the advent of top-down systems biology theory and Whole-Body Systems Biology (WBSB) tools [17].

Table 1. Bioactive compounds in TCHM.

TCHM	Plant Part	Bioactive Compound/s	Medicinal Functions	References
<i>Phyllanthus emblica</i> L.	Air-dried halls	Flavones, Tannins, Polyphenols, Gallic acid, Cinnamic acid, 5-hydroxymethylfurfural, Ellagic acid, Guercetin, β -daucosterol	Anti-diarrhea, Anti-jaundice, Antifungal, Antioxidant, Anti-diabetic, Anti-inflammatory, Anti-tumor Hypolipidemic	[71]
Shanzhuyu (<i>Cornus officinalis</i>)	Fruits	Triterpenoids, Iridoids, Flavonoids Tannins, Organic acids, Polysaccharides	Immune regulation, Anti-hyper-glycaemia, Anti-aging Anti-oxidant, Renal and, neural protection, Anti-diabetic, Nephropathy	[72]
<i>Flos Lonicerae</i>	Flower buds	Flavonoids, Saponins, Organic acids, Iridoid glycosides	Antioxidant, Antiviral, Cytoprotective, Hepatoprotective, Antimicrobial, Anti-inflammatory	[73-75]
<i>Rhodiola rosea</i>	Roots	Flavonoids, Organic acids, Phenylpropanoids	Antioxidant, Adaptogenic Characteristics	[76-77]
<i>Panax notoginseng</i>	Roots	Flavonoids, Amino acids, Saponins, Volatile oils, Polyacetylenes, Polysaccharides, Triacylglycerol	Haemostatic, Immunostimulatory, Anti-inflammatory, Anti-cytotoxic, Antiplatelet	[78-79]

Table (1) contd....

TCHM	Plant Part	Bioactive Compound/s	Medicinal Functions	References
Saffron (<i>Crocus sativus</i> L.)	Leaves Tunics Corms Stigmas Spaths	Phenols, Carotenoids	Antioxidant, Aphrodisiac, Antidepressant, Antitumor, Antiproliferative, Anti-hypolipemic, Anticonvulsant, Anti-hyperglycemic, Anti-hypoglycemic	[80-85]
Danshen (<i>Salvia miltiorrhiza</i>)	Roots	Polysaccharides, Tanshinone I, Tanshinone IIA, Cryptotanshinone	Antioxidant, Immunoregulatory, Antitumor, Cardio-protective, Hepatoprotective	[11]
Rabbiteye blueberry (<i>Vaccinium ashei</i>)	Fruits Leaves	Polyphenols, Anthocyanins	Antioxidant, Antileukemia Antihypertensive, Antiobesity, Antidiabetic, Anti-stroke, Anti-cancer	[86, 87]
Mulberry (<i>Morus alba</i> L.)	Leaves	Flavonoids, Anthocyanins, Alkaloids, Polyphenols	Antioxidant	[30]
Hawthorn (<i>Crataegus pinnatifida</i>)	Fruits Leaves	Procyanidins, Flavonoids, triterpenoids acid	Antimicrobial, Antiviral, Anticancer, Anti-inflammatory	[88-91]
<i>Litchi chinensis</i>	Fruit pericarp	Phenolic acids, Anthocyanins, Flavonoids, Polysaccharides	Antioxidant, Antiviral, Antibacterial, Anti-inflammatory, Anti-carcinogenic	[92]

Table 2. Potency of TCHM by-products in swine production.

TCHM by-Product	Dosage (g/kg)	Measured Parameters	Reported Findings ¹	References
<i>Ginkgo biloba</i> L.	100	Growth performance	Increased BW and ADG; No difference in ADFI	[8]
		Nutrient digestibility	Increased DM, N and GE digestibility	
		Serum biochemical parameters	Increased TP, ALB, ALP and GLU; Decreased BUN level; Increased Hb, TI and TIBC; Increased SOD and GSH-Px; Decreased TG and TCHO; Decreased ALT and AST	
		Immune function	Increased IgG, IgA and IgM; Increased CD2+ and CD4+ cells; Decreased CD8+; Increased CD4+:CD8+ ratio; Increased proportions of B-cells, MHC-I and MHC-II	
Green tea (<i>Camellia sinensis</i>)	5	Growth performance	No effect on WG, FI and FCR	[44]
		Meat composition	No effect on moisture, CP, CF and CA	
		TBA value of meat	No effect	
		Cholesterol of meat	No effect	
		Immunity	Improved cell-mediated immunity; Improved humoral immunity; Increased cytokines IL-6 when stimulated with Con A; Increased cytokines TNF- α when stimulated with LPS	

¹Note: ADFI, Average Daily Feed Intake; ADG, Average Daily Gain; ALB, Albumin; ALP, Alkaline Phosphatase; ALT, Glutamic-Pyruvic Transaminase; AST, Glutamic-Oxalacetic Transaminase; BUN, Blood Urea Nitrogen; BW, Body Weight; CA, Crude Ash; CF, Crude Fibre; Con A, Concanavalin; CP, Crude Protein; DM, Dry Matter; FCR, Feed Conversion Ratio; FI, Feed Intake; GE, Gross Energy; GLU, Glucose; GSH-Px, Glutathione Superoxide Dismutase; HB, Hemoglobin; IgG, Immunoglobulin G; IL-6, Interleukin-6; IgA, Immunoglobulin A; IgM, Immunoglobulin M; LPS, Lipopolysaccharide; MHC-I, Histocompatibility Complex-I; MHC-II, Histocompatibility Complex-II; N, Nitrogen; SOD, Superoxide Dismutase; TCHO, Total Cholesterol; TI, Total Iron; TIBC, Total Iron Binding Capacity; TP, Total Protein; TG, Triglyceride; WG, Weight Gain.

Table 3. Potency of TCHM by-products in poultry production.

TCHM by-Product	Experimental Animal	Dosage g/kg	Measured Parameters	Reported Findings ²	References
Red ginseng marc	Chicken	30	Growth performance	No effect on WG, FI and FCR	[37]
			Serum cholesterol	Reduced total cholesterol, LDL cholesterol and triglyceride; Increased HDL cholesterol	
			Meat composition and quality	No effect on moisture, CP and CA; Reduced CF; Improved antioxidative activity; No effect on WHC and shear force values; No effect on thigh muscle colour	
Pomegranate (<i>Punica granatum</i> L.)	Chicken	20	Growth performance	No significant effect on ADG and ADFI; Reduced FCR	[32]
			Meat composition	Higher moisture; Higher CP; Lower EE	
			Cholesterol	Reduced Cholesterol concentration	
			Trace minerals	No effect on Ca; Higher Fe, Na and Mg	
			Oxidative stability and pH	Decreased TBARS from days 7 to 28 in breast meat and the entire period in thigh meat; Lower pH in breast meat after 14 days of storage	
Oriental herbal medicine ¹	Duck	5	Growth performance	No effect on WG, FI and FCR	[43]
			Serum biochemical parameters	No significant effect on total protein, albumin, aspartate aminotransferase, alanine transaminase and BUN	
			Cholesterol	Reduced Cholesterol concentration	
			Serum antioxidant enzyme activities	Higher serum SOD and CAT activities	
			Meat composition	Increased CP; Reduced CF; Increased S	
			Meat quality	lower PML; Higher WHC; No significant effect on pH and cooking loss; No significant effect on lightness and yellowness; Higher Redness than control	
			Fatty acid composition	Higher palmitic acid, palmitoleic acid, oleic acid, stearic acid and γ -linoleic acid	
			TBARS concentration during cold storage	Decreased TBARS concentration from 15 to 27 days	

¹Note: Oriental herbal medicine composition, 15% velvet Antler, 5% Polydonati rhizome, 10% Astragalus membranaceus, 5% Angelica gigas, 10% Cnidium officinale, 10% Lycium chinense, 10% Atractylodes japonica, 10% Glycyrrhiza uralensis, 10% Cinnamomum cassia, 10% Rehmannia glutinosa, and 5% Hericium erinaceum.

²Note: ADFI, Average Daily Feed Intake; ADG, Average Daily Gain; CA, Crude Ash; CAT, Catalase; CF, Crude Fiber; CP, Crude Protein; EE, Ether Extract; FCR, Feed Conversion Ratio; FI, Feed Intake; HDL, High-Density Lipoprotein; LDL, Low Density Lipoprotein; PML, Percentage Moisture Loss; SOD, Superoxide Dismutase; TBARS, Thiobarbituric Acid Reactive Substances; WG, Weight Gain; WHC, Water Holding Capacity.

Table 4. Potency of TCHM by-products in ruminant production.

TCHM by-Product	Experimental Animal	Dosage g/kg	Measured Parameters	Reported Findings ²	References
TCHM mixture ¹	Cattle	30	Growth performance	Increased ADG and FE; No effect on FI	[34]
			Blood parameters	No effect on RBC and platelets; Higher hemoglobin and hematocrit; Higher plasma albumin; Higher non-esterified fatty acid NEFA	
			Carcass Characteristics	No effect on carcass weight, carcass rate, backfat thickness and rib eye muscle area; Improved meat grade	
Pomegranate	Lactating Cow	40	Performance	Increased FI; Higher ruminal pH; 8.2% increase in average milk production; Lower milk fat; Higher milk protein	[38]
			Digestibility	Higher DM, CP, NDF, cellulose, and hemicellulose digestibility	
Pistachio	Sheep		FI	No effect on DM, OM and CP intake; Decreased NDF intake; Increased EE intake	[36]
			Apparent digestibility	Increased ATTP of CP, EE and OM	
			N balance	Decrease in N excreted; Increase in N retention	
			Blood metabolites	No effect on Serum BUN, glucose, creatinine, potassium and iron	

¹Note: TCHM mixture, *Angelica gigas nakai*, *Rehmannia radix preparata*, *Zingiber officinale roscoe*, *Paeoniae radix*, *Cnidii rhizome*, *Zizyphus jujube*, *Bupleuri radix*, *Citrus nobilis makino*, *Prunus mume* and *Plantago asiatica*.

²Note: ADG, Average Daily Gain; BUN, Blood Urea Nitrogen; CP, Crude Protein; DM, Dry Matter; EE, Ether Extract; FE, Feed Efficiency; FI, Feed Intake; N, Nitrogen; NDF, Neutral Detergent Fiber; NEFA, Non-Esterified Fatty Acid; OM, Organic Matter; RBC, Red Blood Cell.

CONCLUSION

The legal restriction placed on the use of synthetic antibiotics as animal feed additives in some countries has aroused an urgent need for safe and viable alternatives to dietary antibiotics that will improve the health and performance of animals. An alternative source is TCHM by-products because they contain several bioactive compounds that are beneficial to animals. Also, they are biodegradable, less toxic and more environmentally friendly than synthetic antibiotics. This reason, coupled with the increased awareness of the numerous benefits of TCHM, has led to an increase in its production and use. Consequently, the generation of TCHM by-products has increased dramatically, which brings about another concern of environmental pollution. According to the findings of this review, TCHM by-products still contain residual bioactive compounds. This means that instead of discarding TCHM by-products directly, they could be incorporated into animal feed. This is evident in studies that have reported on the benefits of TCHM by-products in animal production. According to the literature cited so far, the effect of TCHM by-products depend on the type of product used as well as the dosage administered. Some reports conclude that TCHM by-products improve animal performance, health, and quality while other reports suggest otherwise. This could be due to the differences in bioactive compounds contained in different TCHM by-products and their respective functions. Generally, TCHM by-products as feed additives, apart from being a good alternative for synthetic antibiotics in animal production, could also minimize the current environmental pollution challenges associated with its disposal. More research needs to be carried out on the potentials of TCHM by-products in animal nutrition, current processing methods, their

mode of action and possible side effects as well as their effect on the environment.

CONSENT FOR PUBLICATION

Not applicable.

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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